

AguaClara Cornell Floc Modeling

Spring 2026: Final Report

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Abstract

The Floc Modeling subteam aims to further our understanding of the mechanics of floc formation and clarification through the development of a mathematical model of these processes. This semester, Floc Modeling is evaluating the distribution of flocs within the clarifier to support or refute the clarification model assumption that floc saturation is distributed uniformly throughout the floc filter. To analyze floc distribution experimentally, we are employing dye tracers to track their formation and movement within the floc filter. In addition, we are revising the current clarification model and its derivations to better describe how primary particles are removed as water travels through the floc filter. The results of these efforts will help evaluate our current understanding of flocculation and clarification, and guide future improvements to both the experimental system and the clarification model.

Introduction

Flocculation and clarification are critical to the AguaClara treatment process as they remove most primary particles before filtration and chlorination. While these processes are widely used throughout the water treatment industry, gaps remain in our understanding of their underlying physical mechanisms. This lack of understanding has made it challenging to optimize the design of treatment plants and develop automated systems that can successfully respond to fluctuating conditions. It has also prevented the widespread adoption of the novel floc filter technology used in AguaClara plants during clarification. For these reasons, an accurate mathematical model of these processes is essential.

This semester, the Floc Modeling subteam is focusing its efforts on refining the AguaClara clarification model. Clarification is the process following flocculation in which flocs are separated from treated water through settling, also known as sedimentation. During conventional clarification, sedimentation is the sole mechanism for particle removal. However, AguaClara plants utilize an additional process called floc sweeping. Floc sweeping allows previously formed flocs to capture additional primary particles that would settle too slowly on their own, increasing the overall efficiency of clarification. In order to achieve this, vertical jets of water are used to maintain a suspension of flocs known as the floc filter. Primary particles travel up through this suspension, colliding and aggregating with the suspended flocs, which can then be removed. Smaller flocs travel up towards a set of plate settlers, which aggregate these small flocs into larger ones that eventually settle back into the floc filter.

Floc sweeping efficiency depends on coagulant dose, floc filter concentration, and floc saturation. Flocs are initially very porous; as a floc captures primary particles, its pores fill. Eventually, a floc cannot capture any additional particles. Floc saturation is a measurement characterizing how “full” a floc is and how many more particles it is able to capture. Spatial variations in floc saturation within the clarifier can affect particle removal efficiency in the clarifier. If a mathematical model can predict the spatial distribution of flocs in the clarifier, then the areas of the clarifier with the most saturated flocs can be targeted first during wasting. Thus, only flocs that are still capable of capturing primary particles will remain in the clarifier, increasing clarification efficiency.

Modeling clarifier performance is crucial for determining accurate and efficient coagulant dosages because coagulant directly influences floc formation, floc size, and the rate of particle removal. Coagulant doses that are too low fail to neutralize particles, and primary particles fail to aggregate, leading to poor floc formation and higher effluent turbidity. In contrast, coagulant doses that are too high can cause charge reversal, causing particles to repel each other and remain suspended. Additionally, the excess coagulant contributes to increased sludge production and higher operation costs. Both overdosing and underdosing coagulants lead to poor floc formation, higher operating costs, and longer treatment times. Therefore, to maximize performance and reduce costs, it's crucial to model clarifier performance in order to determine the correct coagulant doses.

Literature Review & Previous Work

Previous Work

In Spring 2025, the team revised the existing continuous model for floc filter clarification. The model operates in two stages: it first uses the AugaClara Hydraulic Flocculation model to obtain the concentration of primary particles following flocculation, and then models the interactions between these primary particles and flocs in the floc filter in order to determine the concentration of primary particles following clarification. The clarification portion of the model is shown below in Equations 1 and 2.

$$C_{clarified} = C_{flocculated} \exp\left(-k_c \alpha_c h_{floc\ filter}\right) \text{ where } k_c = \frac{\pi r_{floc}^2 C_{floc\ filter}}{m_{floc}} \quad (1)$$

$$\alpha_c = 1 - \left(1 - k' \frac{C_{coag}}{C_{p,0}} (1 - P_{sat})^{2/3}\right)^{n_c} \quad (2)$$

One of the main revisions to the model made by the Spring 2025 team was the introduction of n_c in the calculation for the attachment efficiency α_c . n_c accounts for multiple internal collisions occurring between a primary particle and a single floc since flocs are highly porous and much larger than primary particles, meaning a primary particle entering a floc is likely to run into many potential attachment sites. Furthermore, the model factors coagulant coverage into the calculation of attachment efficiency by using a constant, k' , that converts the

ratio of coagulant concentration to primary particle concentration into coagulant coverage. The expression for k' is given in Equation 3. Note that the equation in the original report was

incorrectly scaled by $\frac{r_{coag}^3}{r_{clay}^3}$.

$$k' = \frac{\rho_{clay} r_{clay}}{4\rho_{coag} r_{coag}} \quad (3)$$

In Fall 2025, the team investigated the use of dye tracers to visualize floc behavior within the mini clarifier. The two dyes evaluated were Red-40 and methylene blue. Red-40 was tested first because it is an anionic dye that could potentially adsorb onto the positively charged PACl. Jar tests were conducted to determine the concentrations of clay, coagulant, and dye that would produce distinguishable flocs without too much residual dye in solution. Results showed that while Red-40 could produce visible flocs, a significant amount of the dye remained dissolved in solution even at the lowest concentrations that still made visible flocs. This residual dye adsorbed onto other flocs within the system, which caused the entire floc filter to be colored red and made it impossible to distinguish tracer flocs from non-tracer flocs, even when flocs did still form successfully.

Because of this limitation, methylene blue was evaluated as an alternative tracer dye. Methylene blue is a cationic dye that adsorbs strongly onto the negatively charged kaolin clay particles. As such, the methylene blue was mixed with clay before adding PACl during jar tests, effectively making ‘methylene blue-dyed clay particles’ that are turned into flocs instead of first making flocs that are then dyed with Red-40. In these experiments, the methylene blue dye strongly attached to the clay and was substantially removed from solution during floc formation. As a result, the flocs formed were visually distinguishable while the surrounding water remained mostly clear. Based on these observations, it was concluded that methylene blue was the more viable tracer dye to use for future experiments.

During the Fall 2025 experiments there were several issues identified in the existing mini clarifier design. The original clarifier had sampling ports along its side and a floc hopper with a flat bottom. Over time, the mini clarifier also began to leak significantly, which reduced the reliability of the experiments performed. To address these issues, several revisions were made to the mini clarifier.

First, the side sampling ports were removed to reduce potential leakage points. Second, a lip was added around the clarifier body to improve the seal between the acrylic and the main body. Third, the floc hopper was modified to have a downward-sloping bottom to help sludge move toward the waste outlet. These modifications were incorporated into a new clarifier design for future experiments.

Floc Filter Imaging

Hurst et al. (2014) developed a methodology for measuring suspended solids concentration within a floc blanket using image analysis, providing a non-destructive method

that captures spatial and temporal variations in floc concentration across the entire reactor. In their setup, a backlit reactor and camera system were used to measure the attenuation of light transmitted through the suspension, which was then converted into local solids concentration using calibrated relationships between light attenuation and particle concentration.

This approach allowed concentration measurements to be obtained for each pixel of the image, enabling high-resolution mapping of the floc blanket structure rather than relying on discrete grab samples. As a result, image analysis offered significantly improved spatial and temporal resolution in characterizing suspended solids dynamics within the reactor.

This methodology directly informed the approach used by our subteam to analyze the spatial distribution of flocs and evaluate the assumption that floc saturation is uniform throughout the floc filter. Rather than measuring solids concentration directly as in Hurst et al. (2014), our work adapts the same imaging principle to observe the movement and distribution of tracer flocs within the clarifier. By tracking dyed flocs within the floc filter, the experiment aims to qualitatively and eventually quantitatively assess how flocs are distributed vertically within the clarifier and whether saturation varies with height. This adaptation builds on the precedent established by Hurst et al. by using imaging as a tool to capture spatial variability in floc behavior, but shifts the focus from measuring concentration to evaluating the uniformity of floc saturation within the clarifier.

Clarification Modeling

Swetland et al. (2014) provided an initial approach to modeling flocculation-sedimentation performance based on coagulant coverage. They derived Equation 4 to describe the average fractional coagulant coverage of primary particles in terms of the number of precipitated coagulant nanoparticles per primary particle, $N_{per\ clay}$, and the ratio of clay surface area to total surface area in the flocculator, R_{clay} . $N_{per\ clay}$ provides an estimate for how much coagulant is available for each primary particle, and R_{clay} accounts for the loss of coagulant being adsorbed onto the walls of the flocculator, assuming coagulant particles have equal affinity towards the flocculator walls and clay particles. A Poisson distribution was used to account for the effect of overlapping coagulant particles reducing the total fractional coverage. Equations 4, 5, and 6 show the formulas for Γ , $N_{per\ clay}$, and R_{clay} , respectively.

$$\Gamma = 1 - \exp\left(-\frac{4r_{coag}^2}{SA_{clay}}N_{per\ clay}R_{clay}\right) \quad (4)$$

$$N_{per\ clay} = \frac{(C_{coag} - C_{coag(aq)})V_{clay}\rho_{clay}}{\frac{4}{3}\pi r_{coag}^3\rho_{coag}C_{P,0}} \quad (5)$$

$$R_{clay} = \frac{SA_{clay\ total}}{SA_{clay\ total} + SA_{wall}} \quad (6)$$

Swetland et al. combined their expression for fractional coagulant coverage with the velocity gradient of the flocculator, G , the residence time, θ , and the average time between

collisions, which they determined to scale with $\phi^{-\frac{2}{3}}$, where ϕ denotes the volume fraction of flocs in the flocculator. From this, they obtained the dimensionless parameter $G\theta\phi^{2/3}$ as a measure of the successful collision potential for laminar flow flocculators. Plotting this parameter against the particle removal efficiency, pC^* , indicated the parameter successfully captured the successful collision potential of the flocculator. Equation 7 shows the definition of pC^* .

$$pC^* = -\log\left(\frac{C_{settled}}{C_{influent}}\right) \quad (7)$$

Pennock et al. (2018) linked hydraulic flocculation to the settled effluent turbidity using two different logarithmic models connecting the influent and effluent turbidity of laminar and turbulent flows. When flocculation occurs in turbulent flow, the relative velocities between primary particles are controlled by inertial forces, but when the flow is laminar, viscous forces control the relative velocities between particles. Both models lead to logarithmic equations relating influent turbidity to effluent turbidity, but the viscous flocculation model is preferred due to the widespread use of $\bar{G}\theta$ as opposed to $\varepsilon^{-1/3}\theta$. Additionally, in the 2018 paper, k is a fitted parameter that is determined experimentally to combine different variables like mixing efficiency, particle collision frequency, and flocculation time into one parameter.

Pennock et al. (2023) provided a more sophisticated model for coupled flocculation-sedimentation performance by expanding upon the AguaClara Hydraulic Flocculation model. Equation 8 shows the viscous version of the AguaClara Hydraulic Flocculation model as defined by Pennock et al. (2018).

$$C_p = \left(\frac{2\pi k \bar{\alpha} \bar{G}\theta}{3} (\rho_p \frac{\pi}{6})^{-2/3} + C_{p,0}^{-2/3} \right)^{-3/2} \quad (8)$$

Here, $C_{p,0}$ is the influent concentration of primary particles and C_p is the concentration of primary particles following sedimentation. $\bar{\alpha}$ describes the average attachment efficiency of collisions between suspended particles based on their coagulant coverage. For a pair of interacting particles, the attachment efficiency is the probability that there is a coagulant nanoparticle at the point of contact, which is given by Equation 9 (Pennock et al., 2018).

$$\bar{\alpha} = 1 - (1 - \Gamma)^2 \quad (9)$$

k is a fitted parameter that is calculated experimentally describing the rate of conversion from nonsettleable to settleable particles. Pennock et al. expended on this model by explicitly relating k to clarifier characteristics. Based on experimental data, they rewrote k in terms of the sedimentation capture velocity of the clarifier, the initial terminal velocity of primary particles, and an additional fitted constant, as shown in Equation 10.

$$k = k_0 \exp\left(\kappa\left(1 - \frac{v_c}{v_{t,0}}\right)\right) \quad (10)$$

Substituting Equation 10 into Equation 8 yields the completed Coupled Flocculation-Sedimentation model, shown in Equation 11.

$$C_P = \left(\frac{2}{3}\pi k_0 \exp\left(\kappa\left(1 - \frac{v_c}{v_{t,0}}\right)\right) \tilde{\alpha} \tilde{G} \theta (\rho_P \frac{\pi}{6})^{-2/3} + C_{P,0}^{-2/3}\right)^{-3/2} \quad (11)$$

Since flocculation performance is strongly dependent on the characteristics of the chosen clarification process, linking these processes together makes the model more useful for characterizing overall plant performance.

A key difference between the Coupled Flocculation-Sedimentation model and the Spring 2025 Floc Filter Clarification model is how particle removal occurs. In the Coupled Flocculation-Sedimentation model, particle removal occurs due to differences in particle settling velocity relative to the upward flow velocity. Particles that settle faster than the capture velocity are removed as they come in contact with and slide down the plates. The Spring 2025 Floc Filter Clarification model, on the other hand, focuses on the interactions between primary particles and the floc filter. As primary particles travel up the clarifier, they collide with the suspended flocs, causing those primary particles to attach to the flocs and be removed.

Fluidized Beds

Richardson and Zaki (1954) developed an expression for the drag force on a spherical particle in a suspension of particles. The paper examined the sedimentation of a uniformly dispersed suspension of spherical particles settling at a constant velocity, in which resistance arose solely from viscous forces.

The Stokes drag force equation (Equation 12) describes the drag force on a single spherical particle moving relative to a fluid, where d is the particle diameter, μ is the fluid viscosity, and V is the relative velocity.

$$F = 3\pi\mu Vd \quad (12)$$

However, in sedimentation processes, particles are suspended among many other particles; the effects of the surrounding particles and the container walls must be accounted for. Thus, Richardson and Zaki modeled the suspension as a single spherical particle in a hexagonal cell, with the cell radius R inversely proportional to the particle concentration C . Boundary conditions were applied such that the relative velocity at the particle surface equaled zero and the velocity gradient at the cell surface was zero. The boundary condition on the velocity gradient implied that as the particle concentration increased, the cell size decreased, resulting in a steeper velocity gradient near the particle surface and thereby increasing viscous drag.

Integrating the shear stress over the particle surface yielded the drag force equation shown in Equation 13, in which β_t was a correction factor ($\beta_t = 2.335$) accounting for the presence of other particles in the suspension.

$$F = 3\pi\mu V d \beta_t \quad (13)$$

Richardson and Zaki discussed two configurations for the cell surrounding a particle: first, the horizontal and vertical distances between particles were both equal to $2R$; second, the particles were separated by a horizontal distance of $2R$ but a vertical distance of 0 . The first configuration described the ratio of the particle radius b to the cell radius R in terms of the volumetric concentration. The equation was rearranged for R , yielding Equations 14 and 15, where b denotes the particle radius.

$$\frac{b}{R} = 1.126 C^{1/3} \quad (14)$$

$$R = \frac{b}{1.126 \left(\frac{\text{total volume of particles}}{\text{volume of suspension}} \right)^{1/3}} = \frac{b}{1.126 \left(\frac{\# \text{ of particles} \times (4/3) \pi b^3}{\text{volume of suspension}} \right)^{1/3}} = \frac{(\text{volume of suspension})^{1/3}}{\frac{4}{3} \pi (1.126) [\# \text{ of particles}]} \quad (15)$$

The definition of R was compared to the discretized floc filter definition of separation distance, shown in Equation 16, where m_{floc} was the mass of a floc and C_{floc} was the concentration of flocs.

$$\Lambda = \left(\frac{m_{\text{floc}}}{C_{\text{floc}}} \right)^{1/3} = \frac{(m_{\text{floc}})^{1/3}}{\left[\frac{\# \text{ of flocs} \times m_{\text{floc}}}{\text{volume of floc filter}} \right]^{1/3}} = \frac{(\text{volume of floc filter})^{1/3}}{(\# \text{ of flocs})^{1/3}} \quad (16)$$

Thus, the discretized floc filter model matches Richardson and Zaki's definition of the distance between particles within a constant factor.

Clarification Characteristics

In addition to reviewing specific clarification models, we also reviewed Hurst's (2010) thesis in order to consider the general design characteristics which affect clarifier performance. Hurst found that optimal performance was achieved within a narrow upflow velocity range, where particle removal was maximized; outside this range, efficiency declined due to either floc washout at high velocities or channeling and poor fluidization at low velocities.

Performance improved significantly with stronger upstream flocculation. Higher energy dissipation rates and longer residence times produced larger, denser, and more resilient flocs, which enhanced turbidity removal and reduced dependence on downstream processes. In cases without prior flocculation, the system relied more heavily on increased blanket depth to achieve comparable removal (Hurst 2010).

Alum dosage improved performance up to an optimal level, beyond which additional dosing had minimal effect, indicating a relatively broad operating window. An empirical relationship between influent turbidity and coagulant dose was established, showing predictable scaling behavior across a wide turbidity range (Hurst 2010).

Floc blanket height had a strong influence on removal efficiency. Increasing height enhanced particle capture through greater particle-particle interactions and filtration-like mechanisms, with near linear improvements up to approximately 45 cm. Beyond this threshold, gains diminished, likely due to floc breakup and system limitations (Hurst 2010).

Solids concentration within the floc blanket stabilized under steady-state conditions and increased as upflow velocity decreased, influencing energy dissipation and collision dynamics within the blanket (Hurst 2010).

The addition of lamellar sedimentation significantly improved overall system performance by capturing smaller particles that escaped the floc blanket and recycling solids back into the system, enabling effluent turbidity to fall below 1 NTU under optimized conditions (Hurst 2010). This indicates that despite the importance of the floc filter in AguaClara clarifiers, the plate settlers are still an important component as well and may need to be factored in to the design of our clarification model.

Part I: Experimentation

Methods

Experimental Apparatus

For our experiments, we used a tube flocculator paired with the low hydraulic residence time model clarifier developed by the Summer 2025 team. The tube flocculator had a radius of curvature of 42 mm and an inner diameter of 4.381 mm. Based on calculations described by Tse et al. (2011), at a flow rate of 1.2 mL/s, this gave a velocity gradient of $G = 145 \text{ Hz}$. This flow rate also gave a hydraulic residence time of $\theta = 304 \text{ seconds}$.

The parameters of the clarifier were chosen to mimic the characteristics of an actual AguaClara clarifier. The cross-sectional area of the clarifier was designed to be 1200 mm^2 in order to achieve an upflow velocity of 1 mm/s given a flow rate of 1.2 mL/s since this is the upflow velocity used in AguaClara plants (Weber-Shirk, n.d.). The average initial velocity at the jet was 17 mm/s based on the cross-sectional area of the jet inlet, which was 71 mm^2 . This larger initial velocity ensured the jet would have sufficient momentum to keep the floc filter suspended.

For the plate settlers, a spacing of 1 cm between plates was chosen to avoid floc rollup. Using the equation found in the AguaClara textbook (Weber-Shirk, n.d.), we calculated the necessary plate settler length to be 14.67 cm to ensure the AguaClara standard capture velocity of 0.12 mm/s . However, due to limitations in print size, we chose to reduce the length to 7.33 cm , resulting in a capture velocity of 0.21 mm/s . A plate width of 2.0 mm was chosen to balance the integrity of the print with having the thinnest plates possible to reduce the increase in upflow velocity through the plate settlers. The plates were placed at a 60° angle based on AguaClara clarifier specifications (Weber-Shirk, n.d.). The area above the plate settlers was left open to the air to prevent air bubbles from being trapped in a closed system.

During operation, coagulant and a turbidity source (typically white kaolin clay) were first added to the influent of the tube flocculator. After exiting the flocculator, the newly formed flocs were directed into the clarifier. The jet reverser maintained these flocs in suspension in order to

form a floc filter within the clarifier. Excess flocs were removed via a floc hopper, which drained flocs from the floc filter once they reached a height of 124 mm above the diffuser.

Apparatus Revisions

To improve the adhesion of the front acrylic plate and the clarifier body, a lip was added, as seen in Figure 1. The lip protrudes from the edges of the clarifier body to ensure that the acrylic plate sits flush with the clarifier top and bottom. This adjustment reduces the potential for leakage. Additionally, the floc hopper floor was extended to slope downwards. This encourages wasted sediment to flow downwards into the waste port and prevents clogging and buildup of flocs in the floc filter. Sampling ports were removed due to difficulty in use and the possibility of leakage.

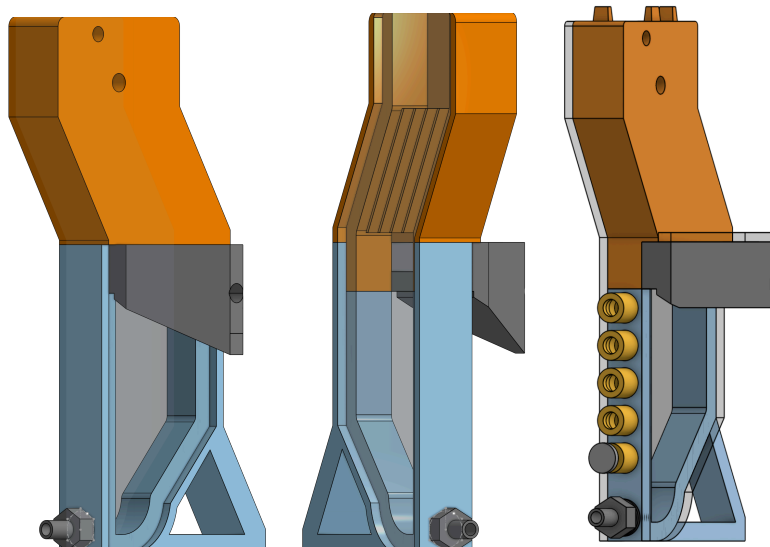


Figure 1. Downward sloping floc hopper and sample ports removed in revised clarifier (left), lip added to clarifier body surrounding acrylic plate (middle), and old clarifier with sampling ports, flat-bottomed floc hopper, and no lip (right).

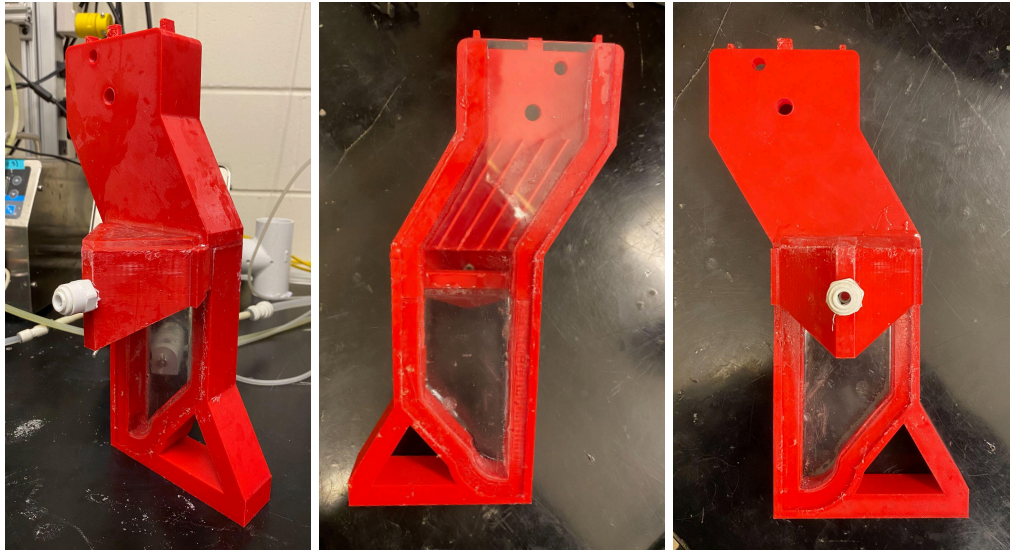


Figure 3. The revised mini clarifier.

Experiment Design

A methylene blue stock solution will first be prepared by dissolving a fixed ratio of 13 mg of methylene blue powder per 1000 mg of deionized water (Ghosh and Bhattacharyya, 2002). A measured volume of this dye stock will then be added to a kaolin clay suspension and mixed thoroughly before addition of coagulant. This order of addition is important because methylene blue is expected to adsorb onto the negatively charged clay particles before PACl is introduced. As a result, the clay particles will be dyed before floc formation, so that adding PACl will form methylene blue-dyed flocs rather than attempting to dye already-formed flocs.

After the methylene blue and clay suspension is prepared, PACl will be added to initiate flocculation. Initial tests will be performed as jar tests to screen combinations of methylene blue-dyed clay and PACl concentrations before running the full tube flocculator and mini clarifier system. These jar tests will be used to identify conditions that produce visible blue flocs while minimizing excess dissolved dye in the water. The main observations recorded will include floc visibility, approximate floc size, settling behavior, and whether dye appears to seep into the surrounding solution.

During the actual mini clarifier experiments, a helical tube flocculator will be used to produce uncolored flocs with kaolin clay. Concentrations of clay and coagulant are subject to change based on experimental results. At a 1.2 mL/s influent rate, the clay will be introduced into the flocculator at a base clarifier influent concentration of 50 mg/L from a 1300 mg/L stock solution, along with 20 mg/L PACl from a 648 mg/L stock solution. Once a steady turbidity is reached, the floc filter will be allowed to reach steady state.

Once the floc filter has reached steady state, the white kaolin clay in the influent will be replaced with the methylene-blue-dyed clay suspension, and PACl will continue to be added to the solution, forming blue dyed flocs as it passes through the flocculator. After exiting the flocculator, the newly formed dyed flocs will enter the clarifier, where the jet reverser will

maintain the flocs in suspension. The distribution of the dyed tracer flocs will then be observed through the clear acrylic face of the clarifier.

Table 1 shows proposed concentrations of PACl and blue-dyed clay concentration to be used in experimentation. Because methylene blue will be mixed with kaolin clay before PACl addition, the table reports the tracer dose as the final concentration of blue-dyed clay in the clarifier influent. The methylene blue dose within the clay stock will be the same for all trials.

Table 1. Proposed Concentrations of Coagulant and Dyed Clay for Experimentation

Trial	Blue-dyed clay concentration	PACl concentration	Purpose
1	30 mg/L	10 mg/L	Standard from Summer 2025
2	10 mg/L	10 mg/L	Test whether lower tracer concentration still produces visible dyed flocs
3	40 mg/L	10 mg/L	Test whether higher tracer concentration improves visibility and floc filter density
4	30 mg/L	5 mg/L	Test whether lower PACl still forms stable dyed flocs
5	30 mg/L	20 mg/L	Test whether higher PACl improves aggregation
8	30 mg/L	0 mg/L	Control to check whether dyed clay remains suspended without flocculation
9	0 mg/L	10 mg/L	Control to observe PACl behavior without clay particles
10	30 mg/L undyed clay	10 mg/L	No-dye control for normal white floc behavior

See Appendix Section IV for Summer 2025 reference.

The initial analysis will focus on whether the tracer flocs are visually distinguishable within the floc filter. Images or videos will be recorded to track where the dyed flocs accumulate, whether they remain suspended, and whether they move uniformly throughout the clarifier. Then, image analysis could be used to quantify spatial variations in the tracer floc distribution and evaluate whether floc saturation appears uniform throughout the floc filter. One method includes converting images to RGB stacks and stripping red and green values (depending on dye color used). The blue image will then be converted to an HSB stack, where saturation values will be read. The average saturation of each row and column of the floc filter will be determined, and

if saturation values remain constant regardless of position, the floc filter will be deemed well-mixed and therefore uniformly distributed.

Discussion

These methylene-blue-dyed floc experiments have yet to be performed with the new clarifier. They will be the primary focus of future experimental work next semester.

Part II: Clarification Modeling

Methods

Coupled Flocculation-Sedimentation Model

As described in the Literature Review section on clarification modeling, the Coupled Flocculation-Sedimentation model developed by Pennock et al. (2023) models clarification performance via primary particle capture by plate settlers. However, it does not account for interactions between primary particles and the floc filter, which play a significant role in the efficiency of AguaClara plants. As such, this model was used as a baseline to compare against the Floc Filter Clarification model, which does account for floc filter interactions.

Discretized Model

The Spring 2025 Floc Filter Clarification model described in the Previous Work section utilized a continuous model in its derivation; that is, it modeled the concentration of primary particles in the floc filter as a continuous gradient. However, some possible errors were discovered in the underlying mathematical reasoning behind the model. For this reason, a new model was developed using a discretized approach in order to reduce complexity and compare with the previous model.

Floc filter clarification was initially modeled by dividing the floc filter into horizontal slices along its full height. However, this approach assumes that the concentration of primary particles is horizontally uniform at each particular height. In actuality, primary particle concentration can vary significantly from one side of the clarifier to the other.

Thus, the floc filter clarification model was derived by visualizing the floc filter as a grid of boxes, with each box representing a discrete cube anchored by a single floc. The full derivation is shown [here](#); see Appendix Section III for variable definitions. The main assumption is that the path of primary particles is predominantly straight upwards, with negligible horizontal displacement relative to the floc separation distance.

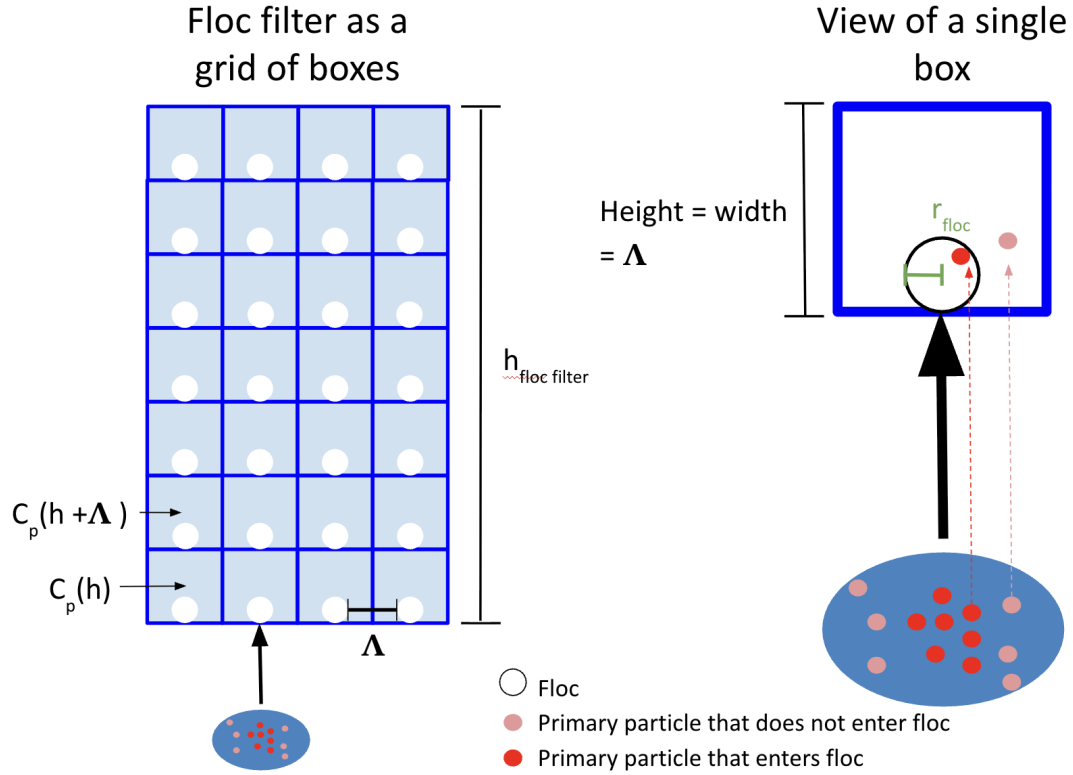


Figure 4. Diagram of the discretized floc filter model.

The concentration of primary particles removed from a single box is defined in Equation 1. The mass flow of primary particles to the floc can be visualized as follows in Figure 4: the floc area in the box can be projected along the box height, creating a cylinder of volume $\pi r_{\text{floc}}^2 \Lambda$. Thus, during the time that it takes for a primary particle to travel the height of the box, the mass flow of primary particles to the floc is simply the number of primary particles that can fit in that cylinder.

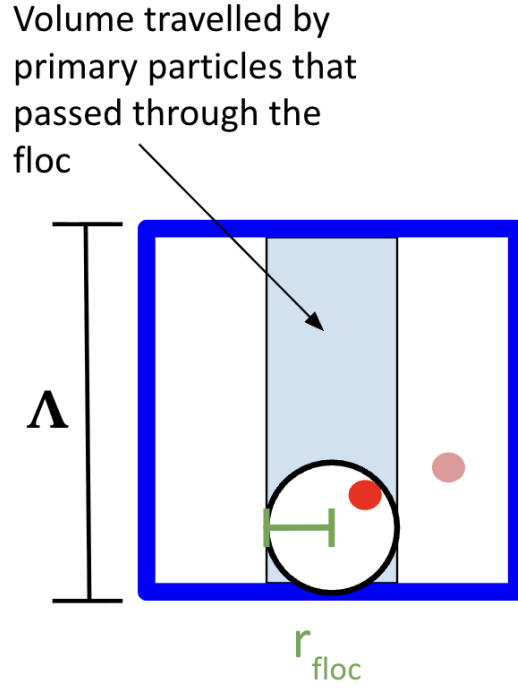


Figure 5. Volume of primary particles through one floc.

$$C_{removed} = \frac{[attachment\ efficiency][Mass\ flow\ of\ primary\ particle\ to\ floc][\#\ of\ flocs\ in\ box]}{[Volume\ of\ box]} = \frac{\alpha \pi r_{floc}^2 C_p(h)}{\Lambda^2} \quad (17)$$

A box at height h has a primary particle concentration of $C_p(h)$ and removes $C_{removed}$ concentration of primary particles. Thus, the next box directly above, at height $h + \Lambda$, has a concentration of primary particles shown in Equation 18.

$$C_p(h + \Lambda) = C_p(h) - C_{removed} = C_p(h) \left(1 - \frac{\alpha \pi r_{floc}^2}{\Lambda^2} \right) \quad (18)$$

Equation 18 was simplified by rewriting Λ in terms of C_{floc} and m_{floc} . A further simplification was made by grouping C_{floc} , m_{floc} , and r_{floc} terms into the floc volume fraction ϕ_{floc} , yielding Equation 19.

$$C_p(h + \Lambda) = C_p(h) \left[1 - \alpha \pi \left(\frac{3\phi_{floc}}{4\pi} \right)^{2/3} \right] \quad (19)$$

By repeating this process for a column of boxes across the entire height of the clarifier, we get Equation 20 from applying a geometric series. Equation 20 relates the concentration of primary particles before clarification, $C_{p,flocculated}$, to that after clarification, $C_{p,clarified}$.

$$C_{p,clarified} = C_{p,flocculated} \left[1 - \alpha \pi \left(\frac{3\phi_{floc}}{4\pi} \right)^{2/3} \right]^{h_{floc\ filter}/\Lambda} \quad (20)$$

When modeling the attachment efficiency, α , the term was split into three multiplicative components to account for the effects of coagulant coverage, floc saturation, and fluid deflection around flocs, as shown in Equation 21.

$$\alpha = \alpha_{div} \alpha_{sat} \alpha_{coag} \quad (21)$$

α_{div} was incorporated to account for the effect of fluid diverging around a floc. When fluid diverges around a floc, the total amount of fluid and primary particles flowing through the floc is smaller than the amount that would have passed through without a floc. As a result, the fluid deflection around a floc decreases overall removal efficiency.

α_{sat} accounts for the effects of floc saturation on attachment efficiency. Equation 22 gives α_{sat} in terms of floc saturation, P_{sat} , and a coefficient relating saturation to attachment efficiency, β . This matches the Spring 2025 model. Note that if α_{sat} is viewed as the probability that a given attachment site is free, then it must be less than one, and as such, β must also be less than one, assuming a linear relationship.

$$\alpha_{sat} = \beta(1 - P_{sat})^{2/3} \quad (22)$$

Floc saturation was determined by assuming a steady state between incoming and outgoing primary particles, similar to the Spring 2025 model. This gives Equation 23.

$$P_{sat} = \frac{C_{p, flocculated} - C_p}{q(C_{p,0} - C_p)} \quad (23)$$

α_{coag} takes into account coagulant coverage and the probability that there will be a coagulant nanoparticle at the site of attachment. This term is very similar to $\bar{\alpha}$ in the Coupled Flocculation-Sedimentation model. However, rather than using Equation 2 for $N_{per\ clay}$, we used a modified form to account for coagulant lost to DOM. We also assumed that clay particles are spherical, allowing us to convey the size of both coagulant nanoparticles and clay primary particles in terms of their radii. This gives us Equation 24.

$$N_{per\ clay} = \frac{(C_{coag} - C_{coag(ag)} - C_{coag(DOM)})r_{clay}^3 \rho_{clay}}{r_{coag}^3 \rho_{coag} C_{p,0}} \quad (24)$$

Furthermore, rather than using a Poisson distribution to account for overlapping areas between coagulant nanoparticles, we used a direct formula for the overlap. An example of overlapping coagulant nanoparticles can be seen in Figure 6. Equation 25 gives the fractional coverage of a surface with total surface area A_T after being covered by n objects each with projected area A_{proj} . The equation was adapted from a derivation used to find the projected area

of overlapping flocs in a floc filter ([see derivation here](#)). While Equation 25 does not account for the curvature of a surface, this was assumed to be negligible since the radius of a clay primary particle is much larger than the radius of a coagulant nanoparticle.

$$A_{cov} = A_T [1 - (1 - \frac{A_{proj}}{A_T})^n] \quad (25)$$

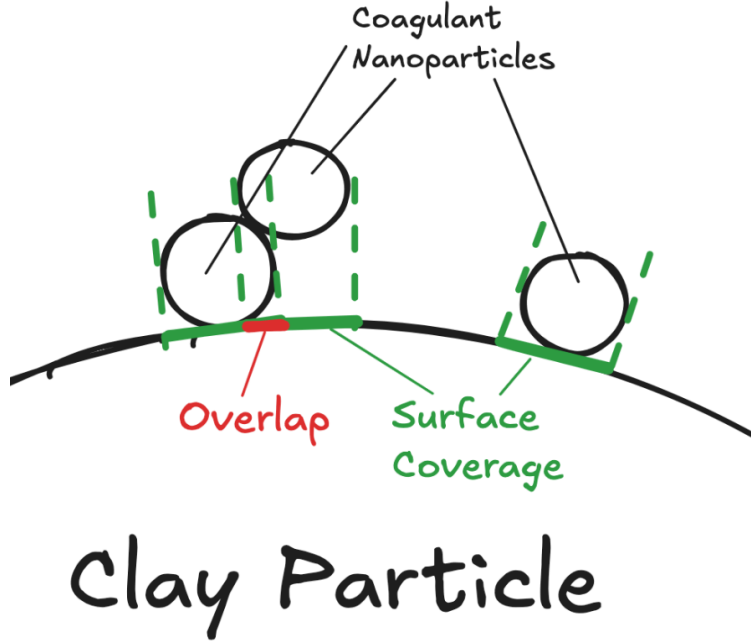


Figure 6. Diagram of overlapping coagulant nanoparticles on a clay particle.

Applying Equation 25 using the projected area of a coagulant nanoparticle, the surface area of a clay particle, and $R_{clay} N_{per\ clay}$ coagulant particles per clay particle yielded Equation 26.

$$\tilde{\Gamma} = \frac{A_{cov}}{A_T} = 1 - (1 - \frac{\pi r_{coag}^2}{4\pi r_{clay}^2})^{R_{clay} N_{per\ clay}} = 1 - (1 - \frac{r_{coag}^2}{4r_{clay}^2})^{R_{clay} N_{per\ clay}} \quad (26)$$

From Pennock et al. (2018), the probability that a floc does not have a coagulant nanoparticle equals $1 - \Gamma$, making the probability that two flocs do not have a coagulant nanoparticle $(1 - \Gamma)^2$. As a result, the probability that one of two flocs has a coagulant nanoparticle on its surface is $1 - (1 - \Gamma)^2$, giving us Equation 27 for α_{coag} .

$$\alpha_{coag} = 1 - (1 - \tilde{\Gamma})^2 = 1 - (1 - \frac{r_{coag}^2}{4r_{clay}^2})^{2R_{clay} N_{per\ clay}} \quad (27)$$

Hence, we have fully defined α . In order to obtain $C_{p, flocculated}$, we can simply apply the AguaClara Hydraulic Flocculation model (Equation 8). However, rather than using $\bar{\alpha}$, we will instead use α_{coag} for our attachment efficiency. We neglect the other two attachment efficiency terms during flocculation since the effect of fluid deflection will be insignificant for similarly sized particles, and floc saturation does not affect floc formation during flocculation.

Note that we could further expand on our formula for α by reintroducing n_c , the number of internal collisions between a primary particle and a floc per interaction between them, using the same structure as the Spring 2025 Clarification model shown in Equation 2. However, this step was excluded to reduce the number of parameters needed to fit the model. n_c was assumed to be equal to 1.

Model Fitting

Both the Coupled Flocculation-Sedimentation model and the Discretized Floc Filter Clarification model were fitted to plant operation data from an AguaClara plant in Nicaragua. Operation data included influent turbidity readings and the corresponding coagulant dosages selected by plant operators. The influent turbidities were first converted to primary particle concentrations based on the ratio of 1 NTU being approximately equivalent to 1.47 mg/L of kaolin clay (Weber-Shirk, n.d.). Each model was then fitted to the operation data under the assumption that the target clarified turbidity was 1 NTU.

For the Coupled Flocculation-Sedimentation model, there were four fitted parameters: A , b , $C_{coag(aq)}$, and $R_{floculator}$. A , b , and $R_{floculator}$ were all aggregated from other unknowns in order to reduce the number of parameters that would need to be fitted. A and b were obtained by rearranging Equation 10 as shown below.

$$k = k_0 \exp\left(\kappa\left(1 - \frac{v_c}{v_{t,0}}\right)\right) = (k_0 e^\kappa) \exp\left(-\frac{\kappa}{v_{t,0}} v_c\right) = A \exp(-b v_c) \quad (28)$$

v_c was estimated to be 0.12 mm/s based on the design parameters for AguaClara clarifiers (Weber-Shirk, n.d.). Note that since we do not know the terminal settling velocity of the primary particles and the capture velocity was the same for all data points, we could also simply fit k to the data directly in order to obtain the same result. This makes the model behave essentially identically to the original AguaClara Hydraulic Flocculation model.

$R_{floculator}$ represents the ratio of the surface area of the flocculator to its volume. This was used to calculate R_{clay} as follows in Equation 29.

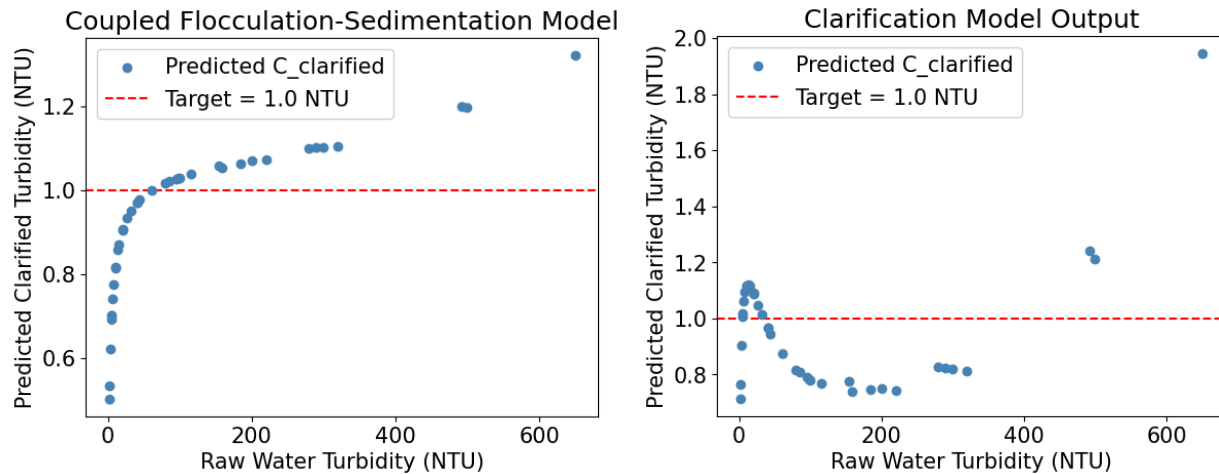
$$R_{clay} = \frac{SA_{clay\ total}}{SA_{clay\ total} + SA_{walls}} = \frac{V_{floculator} N_{clay} SA_{per\ clay}}{V_{floculator} N_{clay} SA_{per\ clay} + SA_{walls}} = \frac{N_{clay} SA_{per\ clay}}{N_{clay} SA_{per\ clay} + R_{floculator}} \quad (29)$$

For the Discretized Floc Filter Clarification model, there were six fitted parameters: k , q , β , Λ , $C_{coag(DOM+aq)}$, and $R_{flocclerator}$. Since β and α_{div} are multiplied together in the model, β was used to account for fitting both parameters. Similarly, since both $C_{coag(aq)}$ and $C_{coag(DOM)}$ are subtracted from the same term, these were also grouped together into one parameter.

After fitting the parameters for each model using the Nelder-Mead algorithm, the fitted models were used to predict the necessary coagulant to obtain an effluent turbidity of exactly 1 NTU at various influent turbidities, which were then compared to the actual plant operators' dosages. This was accomplished by using Brent's method for finding the zeros of a function within a given interval.

Results

Figures 7(a) and (b) show the predicted clarified turbidity at each of the training data points after fitting the models. Note that while the coagulant dose is not shown, both the influent turbidity *and* the coagulant dosage are increasing at each point. Both models show a clear deviation from the 1 NTU effluent target at lower influent turbidities, and both appear to diverge with increasing influent turbidity, though the discretized clarification model appears to diverge more quickly. The Root Mean Squared Error (RMSE) of the Coupled Flocculation-Sedimentation model and the Discretized Clarification model were 0.1867 NTU and 0.2299 NTU respectively, indicating that the coupled model actually performed better than the discretized model. The fitted model parameters are shown in Table 2.

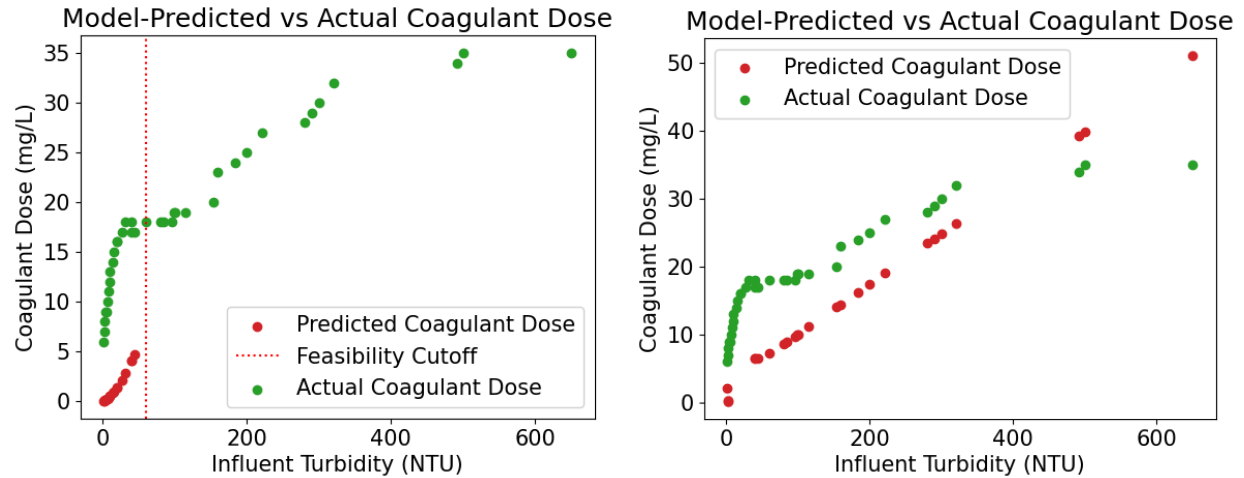


Figures 7(a), 7(b). Graph of predicted clarified turbidity after model fitting for the Coupled Flocculation-Sedimentation model (left) and Discretized Clarification model (right).

Table 2. Fitted model parameters.

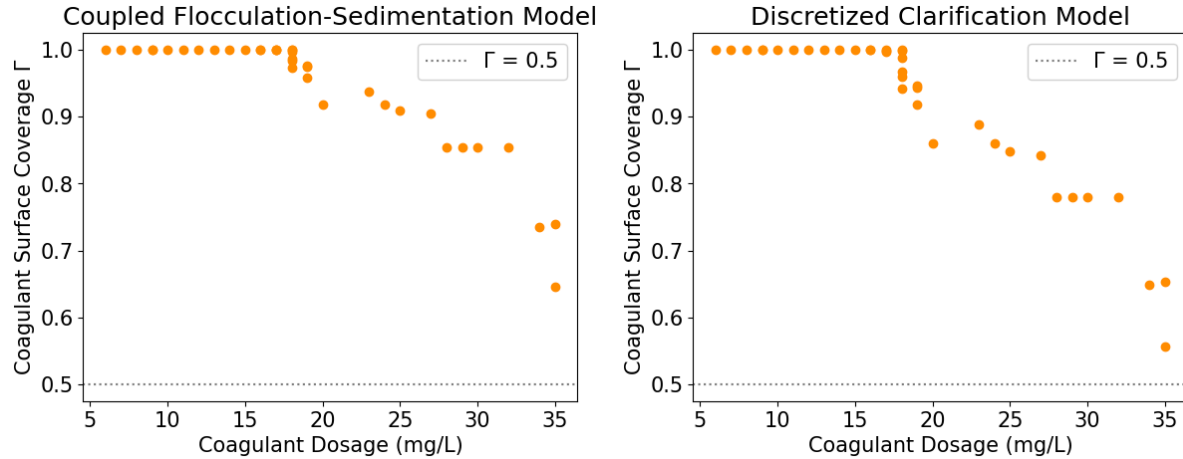
Model Parameter	Coupled Flocculation-Sedimentation Model	Discretized Clarification Model
A	1.506	
b	$2.103 \cdot 10^4$	
C_{DOM}	0	$6.592 \cdot 10^{-6}$
$R_{floculator}$	0.1	0
k		0.00661
q		0.8914
β		$6.7 \cdot 10^{-5}$
Λ		$4.529 \cdot 10^{-6}$

While the Discretized Clarification model appears more erratic and less accurate than the Coupled Flocculation-Sedimentation model in its effluent turbidity predictions, it proved more accurate at predicting coagulant dosages. Figures 8(a) and 8(b) show the predicted coagulant doses needed to reach 1 NTU effluent turbidity according to each model compared with the actual coagulant dosages chosen by plant operators. As shown in Figure 8(a), the Coupled Flocculation-Sedimentation model is unable to predict coagulant dosages for influent turbidities beyond 60.8 NTU. This is because the fitted model predicts that beyond 60.8 NTU, any coagulant dosage will result in an effluent turbidity that is above 1 NTU. The Discretized Clarification model, on the other hand, shows no such cutoff. However, the predicted coagulant dosages are still far from accurate. They appear to form a linear trend that clearly diverges from the actual coagulant dosages as influent turbidity increases.



Figures 8(a), 8(b). Predicted coagulant dosage needed to reach 1 NTU effluent turbidity according to the Coupled Flocculation-Sedimentation model (left) and the Discretized Clarification model (right).

It should be noted that both models predicted extremely high coagulant coverage at lower influent turbidity/coagulant dosage pairs. Figures 9(a) and 9(b) show the calculated coagulant coverages according to both models for each of the data points in the training set. Again, note that although only the coagulant dose values are displayed, *both* coagulant dose and influent turbidity are increasing with each point. These extremely high values for coagulant coverage do not appear physically reasonable. Charge neutralization theory would suggest that such a high coagulant coverage might actually *inhibit* floc formation. Even if this is not the case, it seems unlikely that such a high coagulant coverage would be necessary for optimal performance. This indicates both models are likely missing some key mechanism occurring during flocculation or clarification.



Figures 9(a), 9(b). Average coagulant coverage for each sample in the training data according to the Coupled Flocculation-Sedimentation model (left) and the Discretized Clarification model (right).

Discussion

Clarification Model Improvements

It is clear that more work needs to be done to improve the performance and predictive ability of the Discretized Clarification model. Fixing the extremely high coagulant coverage at lower turbidity/coagulant dose pairs is the most obvious first step. This extremely high coagulant coverage may also be the reason that neither model predicts the correct effluent turbidity at the lower influent turbidities in Figure 7.

One possible explanation is that a large portion of coagulant remains in suspension rather than being adsorbed onto primary particles. The current model assumes that all coagulant that enters the flocculator will eventually adsorb onto either the walls of the flocculator or clay primary particles. However, it is possible that at very low influent turbidities, a large portion of coagulant nanoparticles simply fail to encounter any primary particles on which to adsorb, leaving them stuck in suspension without contributing to the overall coagulant coverage.

Model Evaluation

In addition to modifying the model itself to improve accuracy, it is also possible that our overall methodology for model fitting needs to be revised. The effluent turbidity for each sample likely was not exactly 1 NTU, so this condition may have been imposed too harshly. Obtaining more plant operation data that includes effluent turbidity measurements would obviously be the easiest way to remedy this issue.

Furthermore, it is difficult to tell whether or not the fitted parameters obtained from the optimization are optimal. Due to the large number of parameters in the Discretized Clarification model, global search methods are extremely computationally expensive and time consuming to run, forcing us to use local methods like Nelder-Mead. These algorithms are affected by their initialization position, meaning it may be necessary to run the optimization multiple times from

different feasible initialization positions to ensure the final fit is optimal. It may also be helpful to look for more ways to consolidate multiple parameters into one fitting parameter in order to reduce the dimensionality of the parameter space. This may open the door again for global search methods.

Continuous vs Discretized Clarification Model

The discretized clarification model and Spring 2025 clarification model both describe the concentration of primary particles after clarification in terms of the concentration of primary particles before and other physical constants. Although they use different approaches, the results can be proved to be nearly identical when both models use the same value for α . First, the discretized model from Equation 20 was rewritten into a natural exponential function.

$$C_{clarified} = C_{flocculated} \exp\left[\frac{h_{floc\ filter}}{\Lambda} \ln\left(1 - \alpha\pi\left(\frac{3\phi_{floc}}{4\pi}\right)^{2/3}\right)\right] \quad (30)$$

Then, the Taylor series expansion for $\ln(1 - x)$ was applied to Equation 30, where $x = \alpha\pi\left(\frac{3\phi_{floc}}{4\pi}\right)^{2/3}$ as shown in Equation 31.

$$C_{clarified} = C_{flocculated} \exp\left[-\frac{h_{floc\ filter}}{\Lambda} \sum_{n=1}^{\infty} \frac{1}{n} \left(\alpha\pi\left(\frac{3\phi_{floc}}{4\pi}\right)^{2/3}\right)^n\right] \quad (31)$$

Since the value of $\alpha\pi\left(\frac{3\phi_{floc}}{4\pi}\right)^{2/3}$ was small, the higher-order terms were small enough to be eliminated; thus, a first-order Taylor series approximation was used.

$$C_{clarified} \approx C_{flocculated} \exp\left[-\frac{h_{floc\ filter}}{\Lambda} \alpha\pi\left(\frac{3\phi_{floc}}{4\pi}\right)^{2/3}\right] \quad (32)$$

Substituting in the full expressions for Λ and ϕ_{floc} yielded Equation 33.

$$C_{clarified} \approx C_{flocculated} \exp\left[-\frac{\pi C_{floc} \alpha h_{floc\ filter} (r_{floc})^2}{m_{floc}}\right] \quad (33)$$

Equation 34 shows the Spring 2025 clarification model, which is identical to Equation 33.

$$C_{clarified} = C_{flocculated} \exp\left[-\frac{\pi C_{floc} \alpha h_{floc\ filter} (r_{floc})^2}{m_{floc}}\right] \quad (34)$$

The leading truncation error for Equation 31 was as follows, where $\phi_{floc} = 0.15$ and we assume $\alpha = 1$ for the worst-case error (Hurst, 2010).

$$LTE = \frac{1}{2} \left(\alpha\pi\left(\frac{3\phi_{floc}}{4\pi}\right)^{2/3}\right)^2 \approx 0.027 \quad (35)$$

Nevertheless, the difference between the results of the discretized and Spring 2025 models was minimal, even with LTE included.

$$C_{clarified, discretized} = C_{flocculated} \exp\left[-\frac{h_{floc\ filter}}{\Lambda} \alpha \pi \left(\frac{3\phi_{floc}}{4\pi}\right)^{2/3}\right] \exp\left[\frac{1}{2} \left(\alpha \pi \left(\frac{3\phi_{floc}}{4\pi}\right)^{2/3}\right)^2\right]$$

$$C_{clarified, discretized} = (e^{0.027}) C_{clarified, continuous} = 1.03 C_{clarified, continuous} \quad (36)$$

Thus, the discretized clarification model and Spring 2025 clarification model can be used essentially interchangeably for the same values of α . Since the discretized model uses a power function, whereas the Spring 2025 model uses an exponential function, this results in two equivalent models that use different modeling approaches. As such, if large quantities of data need to be modeled, there are now two models to choose from, based on efficiency and ease of computation, that yield the same result. At the same time, the two models share the issue of unphysical coagulant surface coverage. Thus, it is crucial to conduct further research into the error and correct the models appropriately.

Future Work

The dye methylene blue will be used as a tracer dye in future clarification experiments due to its cationic character and favorable absorption onto raw kaolin clay. Although the optimal dosages of coagulant still need to be determined, successful jar tests with methylene blue from Fall '25 have demonstrated that methylene blue dyed flocs are clearly visible with minimal dye seepage into water. With the newly modified model clarifier, tests for floc saturation will be implemented using methylene blue dyed tracer flocs. Initial analysis of floc spatial distribution will be mainly qualitative, but eventually, image processing and analysis will be used to quantitatively describe the spatial distribution of tracer flocs. This analysis could be performed using ImageJ software to measure the saturation values of individual pixels and determine if the average saturation remains constant along rows and columns of the floc filter.

Additionally, the current model significantly underestimates the actual dosage of coagulant. This seems to be caused by the model predicting extremely high coagulant surface coverage, which is unphysical. Since the Spring 2025 clarification model is equivalent to the current model (by a first order Taylor series approximation), the same error should hold across models. More work needs to be done to identify the causes for the extremely high coagulant coverage and adjust the model accordingly. Fitting this updated model to data acquired from POST will reveal the validity of the model and give insights into where the model can be improved.

Appendix

Section I: Spring 2024 Clarification Model

[Writeup of Vertical Clarifier Floc Sweeping Equation Derivation from SP24](#)

Section II: Spring 2025 Clarification Model

[Writeup of Clarifier Floc Equation Derivation - SP25 and Summer](#)

Section III: Variable Definitions for Floc Filter Model

Λ = floc separation distance

N_{floc} = number concentration of flocs

$N_{per\ clay}$ = number of coagulant aggregates per clay particle

N_{clay} = number concentration of clay

$C_{removed}$ = concentration of primary particles removed

$C_p(h)$ = concentration of primary particles at height h in the clarifier

C_p = concentration of primary particles after clarification

$C_{p,0}$ = concentration of primary particles in influent

$C_{clarified}$ = concentration of primary particles after clarification

$C_{flocculated}$ = concentration of primary particles after flocculation

$C_{settled}$ = settled water clay concentration

$C_{influent}$ = raw water clay concentration

C_{coag} = concentration of coagulant

$C_{coag(DOM)}$ = concentration of coagulant bound to DOM

$C_{coag(aq)}$ = concentration of coagulant that remains in solution

Γ = approximated fractional coagulant coverage via Poisson distribution

$\tilde{\Gamma}$ = true fractional coverage of a suspended particle by coagulant nanoparticles

r_{floc} = radius of a floc

r_{clay} = radius of a clay particle

r_{coag} = radius of a coagulant particle

R_{clay} = ratio of clay surface area to total flocculator surface area

$R_{flocculator}$ = ratio of surface area of walls to volume of flocculator

m_{floc} = mass of a floc

C_{floc} = concentration of flocs in clarifier

Φ_{floc} = volume fraction of floc

α = attachment efficiency

α_{div} = fluid deflection component of attachment efficiency

α_{sat} = floc saturation component of attachment efficiency

α_{coag} = coagulant coverage component of attachment efficiency

α_c = attachment efficiency in continuous clarification model

n_c = number of internal collisions between primary particle and floc per interaction

k = fitted parameter characterizing overall flocculation performance

κ = fitted parameter

k_c = clarification constant

k' = conversion factor from coagulant concentration ratio to surface coverage

$h_{floc\ filter}$ = height of the floc filter

$SA_{clay\ total}$ = surface area of all clay particles in flocculator

$SA_{per\ clay}$ = surface area of a single clay primary particle

SA_{wall} = surface area of flocculator wall

V_{clay} = volume of a clay particle

G = velocity gradient of flocculator

$\tilde{G}$ = average velocity gradient

θ = hydraulic residence time

pC^* = particle removal efficiency

$\bar{\alpha}$ = average attachment efficiency

ε = local energy dissipation rate

v_c = clarifier capture velocity

$v_{t,0}$ = initial terminal velocity of primary particles

ρ_p = density of nonsettable primary particles

ρ_{coag} = density of coagulant

ρ_{clay} = density of clay

d = particle diameter

b = particle radius

μ = fluid velocity

V = relative velocity

β_t = correction factor

P_{sat} = floc saturation

β = constant relating saturation to attachment efficiency

q = mass of primary particles that can be captured per mass of floc

A_{cov} = fractional coverage of a surface with surface area A_T

A_{proj} = projected surface area

Section IV: Summer 2025 Experiment Log

[Summer 2025 Experiment Log](#)

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